

LOW ACRYLONITRILE NBR PATENT LANDSCAPE AND PROCESS ANALYSIS

1. ABSTRACT

This report reviews two primary patents relevant to nitrile butadiene rubber (NBR) synthesis and formulation, with emphasis on low-acrylonitrile composition control and process routes that influence acrylonitrile incorporation. The target material is low-acrylonitrile NBR, generally understood as NBR grades below about 20 wt% acrylonitrile, although the provided patent set includes one process patent focused on high-bound-acrylonitrile NBR and one formulation patent that uses NBR as a component in swellable rubber compositions. The landscape therefore spans both direct copolymerization chemistry and downstream rubber compounding, but only one patent directly addresses monomer feed/composition control during NBR polymerization.

US5770660A discloses a continuous free-radical emulsion copolymerization of acrylonitrile and butadiene using multiple reaction vessels and incremental butadiene addition to maintain a controlled unconverted-monomer ratio. Although the patent is aimed at high bound acrylonitrile NBR, it is technically important because it demonstrates how staged monomer addition, continuous transfer, and emulsion polymerization can stabilize composition distribution and conversion behavior in NBR manufacture. The examples contrast conventional batch, incremental batch, and continuous operation, showing that continuous processing improves tensile strength and environmental resistance relative to batch routes.

US10385647B2 is not a synthesis patent for NBR itself, but it is relevant because it incorporates a nitrile elastomer, including NBR, into a swellable rubber composition with butadiene rubber, sulfur curing agent, silica, and powder super absorbent polymer. The patent emphasizes formulation control, high SAP loading, and curing conditions that preserve mechanical integrity under brine, temperature, and pressure. It also discloses a nitrile rubber elastomer option with Mooney viscosity ML 1+4 (100°C) of about 47 to about 72 and discusses hydrogenated nitrile elastomer variants, which are outside the target but useful as formulation context.

Overall, the evidence suggests that low-acrylonitrile NBR synthesis would most plausibly be achieved by adjusting acrylonitrile feed ratio and maintaining emulsion polymerization conditions that avoid late-stage butadiene depletion. The strongest process lessons are staged monomer addition, controlled conversion, and continuous reactor operation; however, the manifest does not include a patent explicitly optimized for low-acrylonitrile NBR.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US5770660A
- Patent Title: Continuous polymerization process for producing NBR rubber having a high bound ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1998
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a continuous free-radical emulsion copolymerization of acrylonitrile and butadiene in a plurality of reaction vessels. Acrylonitrile and an initial butadiene charge are fed to a first reactor, and additional butadiene is added incrementally in downstream reactors to maintain a controlled unconverted-monomer ratio and produce NBR with improved property uniformity. - Acrylonitrile to butadiene feed ratio: about 80 parts acrylonitrile to 20 parts $\pm 10\%$ butadiene — Claim 1 - Unconverted monomer ratio: about 70 to 90 parts acrylonitrile to about 30 to 10 parts butadiene — Claim 1 - Number of reaction vessels: 5 — Example VII - Acrylonitrile charge: 56 parts — Example VII - Initial butadiene charge: 20 parts — Example VII - Additional butadiene charge: 6 parts — Example VII - Additional butadiene charge: 6 parts — Example VII - Additional butadiene charge: 6 parts — Example VII - Steady-state conversion in first vessel: 24 percent — Example VII - Steady-state conversion in second vessel: 40 percent — Example VII - Steady-state conversion in third vessel: 56 percent — Example VII - Steady-state conversion in fourth vessel: 72 percent — Example VII - Steady-state conversion in fifth vessel: 82 percent — Example VII - Final conversion: 82 percent — Example VII - Bound acrylonitrile content: 48.8 percent — Example VII - Butadiene charge: 21 parts — Example I - Acrylonitrile charge: 54 parts — Example I - Conversion: 50.2 percent — Example I - Bound acrylonitrile content: 59.5 percent by weight — Example I - Butadiene charge: 37 parts — Example II - Acrylonitrile charge: 54 parts — Example II - Conversion: 35 percent — Example II - Conversion: 73.3 percent — Example II - Bound acrylonitrile content: 52.6 percent by weight — Example II - Conversion: 73.6 percent — Example III - Bound acrylonitrile content: 54.9 percent — Example III - Butadiene charge: 46 parts — Example IV - Acrylonitrile charge: 54 parts — Example IV - Conversion: 35 percent — Example IV - Conversion: 65 percent — Example IV - Conversion: 85.2 percent — Example IV - Bound acrylonitrile content: 50.5 percent — Example IV - Acrylonitrile charge: 56 parts — Example V - Butadiene charge: 43 parts — Example V - Initial butadiene charge: 22 parts — Example V - Conversion: 32 percent — Example V - Additional butadiene charge: 8 parts — Example V - Conversion: 52 percent — Example V - Additional butadiene charge: 7 parts — Example V - Conversion: 72 percent — Example V - Additional butadiene charge: 6 parts — Example V - Final conversion: 81.9 percent — Example V - Bound

acrylonitrile content: 50.2 percent — Example V - Final conversion: 86.8 percent — Example VI - Bound acrylonitrile content: 50.6 percent — Example VI

Relevant Experimental Evidence - Example I: Conventional batch copolymerization with 21 parts butadiene and 54 parts acrylonitrile gave 59.5 wt% bound acrylonitrile but only 50.2% conversion. - Example II: Two-increment batch addition improved conversion to 73.3% and produced 52.6 wt% bound acrylonitrile. - Example III: Extending batch polymerization to 73.6% conversion yielded 54.9% bound acrylonitrile. - Example IV: Three butadiene increments raised conversion to 85.2% and gave 50.5 wt% bound acrylonitrile. - Example VII: A 5-vessel continuous process maintained staged conversions from 24% to 82% and produced NBR with 48.8% bound acrylonitrile and improved properties.

Technical Relevance This patent is highly relevant because it directly addresses acrylonitrile/butadiene copolymerization control, the key lever for tuning NBR composition. Although it targets high bound acrylonitrile, the staged-feed continuous emulsion strategy is directly informative for controlling monomer ratio and composition drift in low-acrylonitrile NBR synthesis.

Patent 2

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR synthesis; instead, it discloses swellable rubber compositions containing a nitrile elastomer, butadiene rubber, sulfur curing agent, silica, and powder super absorbent polymer. The nitrile elastomer is compounded and cured into swellable packers designed to expand in brine and withstand high temperature and pressure without excessive leaching or perforation. - Powder super absorbent polymer: about 125 Phr to about 175 Phr — Claim 2 - Powder super absorbent polymer: about 170 Phr to about 210 Phr — Claim 3 - Powder super absorbent polymer: at least about 150 Phr — Claim 4 - Powder super absorbent polymer: about 175 Phr — Claim 4 - Powder super absorbent polymer: about 200 Phr — Claim 4 - Powder super absorbent polymer: about 225 Phr — Claim 4 - Powder super absorbent polymer fraction: at least about 50% by weight of total SAP — Claim 5 - Powder super absorbent polymer fraction: about 60% by weight of total SAP — Claim 5 - Powder super absorbent polymer fraction: about 70% by weight of total SAP — Claim 5 - Powder super absorbent polymer fraction: about 80% by weight of total SAP — Claim 5 - Powder super absorbent polymer fraction: about 90% by weight of total SAP — Claim 5 - Powder super absorbent polymer fraction: about 95% by weight of total SAP — Claim 5 - Mooney viscosity ML

1+4 (100 degrees Celsius): about 47 to about 72 — Synthesis passages - Percent hydrogenation: about 70 percent to about 91 percent — Synthesis passages - Percent unsaturation: about 9 percent to about 30 percent — Synthesis passages - Swelling: at least about 50% by volume in about 96 hours — Synthesis passages - Swelling: at least about 60% by volume in about 96 hours — Synthesis passages - Swelling: at least about 65% by volume in about 96 hours — Synthesis passages - Swelling: at least about 70% by volume in about 96 hours — Synthesis passages - Swelling: at least about 75% by volume in about 96 hours — Synthesis passages - Swelling: at least about 80% by volume in about 96 hours — Synthesis passages - Temperature: about 200 degrees Fahrenheit — Example 3A - Temperature: about 275 degrees Fahrenheit — Example 4A - Brine concentration: about 5% weight/volume — Example 5A - Brine concentration: about 16% weight/volume — Example 6A - Pressure: at least about 3000 psi — Example 9A - Pressure: at least about 4000 psi — Example 9A - Pressure: at least about 5000 psi — Example 9A - Pressure: at least about 6000 psi — Example 9A - Pressure: at least about 70000 psi — Example 9A - Pressure: at least about 8000 psi — Example 9A - Pressure and temperature: at least about 5000 psi and at least about 200 degrees Fahrenheit — Example 11A - Pressure and temperature: at least about 5000 psi and at least about 275 degrees Fahrenheit — Example 12A - Base pipe outer diameter: 4.5 inches — Example 1 - Packer outer diameter: 5.7 inches — Example 1 - Base pipe length: five feet — Example 1 - Curing temperature: 330 degrees Fahrenheit — Example 1 - Curing pressure: 70 psi — Example 1 - Curing time: 2.5 hours — Example 1 - Test temperature: 275 degrees Fahrenheit — Example 1 - Test pressure: about 300 psi — Example 1 - Differential pressure: at least 5000 psi — Example 1 - SAP loading: 110 Phr — Example 2 - SAP loading: 130 Phr — Example 2 - SAP loading: 150 Phr — Example 2 - SAP loading: 170 Phr — Example 2 - Particle size mean: 35.33 μm — Example 2 - Particle size median: 5 μm — Example 2 - Particle size standard deviation: 10.61 μm — Example 2 - Particle size mean: 298.67 μm — Example 2 - Particle size median: 261 μm — Example 2 - Particle size standard deviation: 97.72 μm — Example 2 - Particle size mean: 666.23 μm — Example 2 - Particle size median: 652 μm — Example 2 - Particle size standard deviation: 161.09 μm — Example 2 - Predicted SAP loading for 75% swell using powder SAP: about 205 Phr — Example 3 - Predicted SAP loading for 75% swell using medium granular SAP: about 190 Phr — Example 3 - Maximum load for medium granular SAP: about 170 Phr — Example 3 - Predicted SAP loading for 75% swell using large granular SAP: about 160 Phr — Example 3 - Maximum load for large granular SAP: about 150 Phr — Example 3 - NBR rubber content: 10% — Example 4 - HNBR content: 10% — Example 4 - Swell after 96 hours at 200 degrees Fahrenheit in 5% brine: 75.8% — Example 4 - Swell after 96 hours at 200 degrees Fahrenheit in 16% brine: 46.5% — Example 4 - Swell after 96 hours at 275 degrees Fahrenheit in 5% brine: 98.4% — Example 4 - Swell after 96 hours at 200 degrees Fahrenheit in 16% brine: 65.4% — Example 4 - Swell after 96 hours at 200 degrees Fahrenheit in 5% brine for HNBR composition: 77.8% — Example 4 - Swell after 96 hours at 200 degrees Fahrenheit in 16% brine for HNBR composition: 47.2% — Example 4 - Swell after 96 hours at 275 degrees Fahrenheit in 5% brine for HNBR composition: 96.0% — Example 4 - Swell after 96 hours at 200 degrees Fahrenheit in 16% brine for HNBR composition: 66.9% — Example 4

Relevant Experimental Evidence - Example 1: A 10-foot packer-style swellable rubber article was cured at 330°F and 70 psi for 2.5 hours and then tested in 16% brine at 275°F and about 300 psi. - Example 2: SAP particle size and loading were compared at 110-170 Phr, showing how

powder, medium granular, and large granular SAP affect tensile strength, hardness, leaching, and swell. - Example 3: Powder SAP was predicted to achieve 75% swell at about 205 Phr, while medium and large granular SAP were limited by maximum practical loadings of about 170 and 150 Phr. - Example 4: A composition containing 10% NBR rubber was compared with 10% HNBR, showing similar swell performance under brine and high-temperature conditions. - Example 1A-12A: The claims define operating windows for swelling time, temperature, brine concentration, and pressure resistance in wellbore sealing applications.

Technical Relevance This patent is relevant because it uses a nitrile elastomer, including NBR, as a base component in a high-performance swellable rubber formulation. While it does not teach NBR polymerization, it provides formulation and cure context for how NBR is used in demanding brine/temperature/pressure environments.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US5770660A is a direct NBR copolymerization patent, while US10385647B2 is a downstream compounding patent that uses NBR as one elastomer component in swellable packers.
- US5770660A controls acrylonitrile/butadiene composition by staged monomer addition in continuous emulsion polymerization; US10385647B2 controls performance by formulation design, SAP loading, and curing conditions rather than monomer feed.
- US5770660A explicitly discloses acrylonitrile/butadiene feed ratios of about 80/20 and unconverted-monomer ratios of about 70-90/30-10; US10385647B2 does not disclose NBR monomer composition or polymerization conditions.
- US5770660A uses multiple reaction vessels and incremental butadiene addition; US10385647B2 uses a nitrile elastomer with butadiene rubber, sulfur curing agent, silica, and powder SAP.
- US5770660A reports conversions from 24% to 82% in the continuous example and bound acrylonitrile content of 48.8%; US10385647B2 reports curing at 330°F and 70 psi for 2.5 hours and swell testing at 200-275°F in 5% and 16% brine.
- For low-acrylonitrile NBR synthesis, US5770660A is the more informative process reference because it addresses monomer ratio control directly, even though its target composition is high bound acrylonitrile.

4. CONCLUSION

The patent evidence contains one direct NBR polymerization process and one formulation patent using NBR in swellable rubber articles. For low-acrylonitrile NBR synthesis, the most useful guidance comes from US5770660A: maintain emulsion polymerization with controlled acrylonitrile/butadiene feed, avoid late-stage butadiene depletion, and use staged or continuous addition to stabilize composition. Although the patent is aimed at high bound acrylonitrile, the

same process logic can be inverted to support lower acrylonitrile incorporation by reducing acrylonitrile feed and preserving monomer-ratio control throughout conversion. US10385647B2 is relevant mainly as application context, showing that NBR can be compounded with sulfur, silica, and powder SAP for high-temperature brine service. Based on the evidence, the recommended synthesis direction for low-ACN NBR is controlled emulsion copolymerization with careful acrylonitrile feed management, staged monomer addition, and conversion control to prevent composition drift.

5. REFERENCES

- US5770660A | Continuous polymerization process for producing NBR rubber having a high bound ... | Not disclosed | US | 1998 | <https://patents.google.com/patent/US5770660A/en>
- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>